

POLICY BRIEF

WHERE MATERNAL HEALTH INVESTMENT IS NEEDED MOST IN GHANA

Findings from the Maternal Health Vulnerability Index (MHVI) — Ghana 2026

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Intended audience: Ghana Health Service, Regional Health Directorates, Ministry of Health

The Core Finding

Ghana's 16 regions are not equal in their exposure to maternal health risk. The most vulnerable region — Upper East — has a Maternal Health Vulnerability Index score nearly five times higher than the least vulnerable region, Ahafo. This gap is not explained by any single factor. It is the result of compounding disadvantages in education, poverty, healthcare access, and sanitation that co-occur in the same northern and central regions and reinforce each other.

The Situation

For years, Ghana has collected detailed regional data on maternal health indicators through the Demographic and Health Survey. That data shows — clearly — that women in Upper East, Bono, North East, and Central regions face a substantially higher burden of risk than women in Greater Accra, Ashanti, Eastern, or Ahafo. But that data has remained scattered across separate indicator reports with no single tool to bring it together into a usable picture for planners.

The MHVI was built to change that. It combines eleven evidence-based indicators into one vulnerability score per region, ranks all 16 regions, and identifies what is driving risk in each one. The findings are summarised below.

Where the Risk Is Concentrated

Risk Level	Regions	MHVI Score	Primary Drivers
 HIGH	Upper East, Bono, North East, Central	0.65 – 0.86	No education, poverty, low ANC attendance
 MEDIUM	Northern, Bono East, Upper West, Western North, Savannah, Oti, Volta	0.29 – 0.36	Mixed access and socioeconomic factors
 LOW	Western, Greater Accra, Ashanti, Eastern, Ahafo	0.18 – 0.29	Stronger health systems, higher education, better infrastructure

What Is Driving Vulnerability in High-Risk Regions

The MHVI identifies the top contributing indicators for each region. Across the four high-vulnerability regions, three factors consistently appear as the strongest drivers:

- Low female educational attainment — the single most powerful predictor of high MHVI across all regions. In the highest-risk regions, a large proportion of women aged 6 and above have received no formal education at all.
- High poverty — the percentage of the population in the lowest household wealth quintile is substantially higher in northern and central regions than in the south.
- Low antenatal care attendance — fewer women in high-risk regions are making four or more ANC visits, and fewer are attending their first visit before four months of pregnancy. The barrier is access and awareness, not quality of care once inside the clinic.

These three factors are not independent. They are deeply interlinked. A woman with no education is more likely to be poor, less likely to know the recommended ANC schedule, and less likely to have the resources or mobility to attend a health facility. Addressing one without addressing the others will produce limited results.

Recommendations

1. Prioritise the four high-vulnerability regions for immediate targeted investment

Upper East, Bono, North East, and Central regions carry MHVI scores that place them in a category of urgent need. Resource allocation decisions — ANC outreach staffing, mobile health unit deployment, community health worker assignment — should weight these four regions significantly higher than the national average. The MHVI dashboard provides the evidence base for making these decisions transparently and consistently.

2. Treat female educational exclusion as a health intervention

The data consistently shows that no indicator predicts maternal vulnerability as strongly as the exclusion of women from formal education. Ghana Health Service and the Ministry of Education should treat girls' school enrolment and retention in high-vulnerability regions not as a separate sector concern but as a direct maternal health intervention. Programmes that keep girls in school through adolescence will produce measurable maternal health benefits within a generation.

3. Shift ANC outreach strategy from attendance to early initiation

The data shows that the gap in ANC quality between regions is smaller than the gap in ANC access. The priority for high-vulnerability regions should be reaching women before their fourth month of pregnancy, not improving what happens inside the clinic. Community health workers in Upper East,

Bono, and North East should be specifically resourced and incentivised for first-trimester outreach, with tracking systems that flag women who have not attended by week 16.

4. Use the MHVI dashboard for ongoing monitoring — not just as a one-time finding

The dashboard built by this project allows any health planner to select a region, see its current MHVI score, identify its top vulnerability drivers, and compare it to the national average. It should be used as a living monitoring tool — updated with each new DHS round — rather than treated as a static report. If a region's MHVI score drops after a targeted intervention, that is evidence the intervention worked. If it does not drop, that is a signal to change strategy.

5. Extend the tool to district level

The current MHVI operates at regional level — 16 regions. Within each region there is significant district-level variation that this version cannot capture. Investing in district-level data collection and extending the MHVI to Ghana's 261 districts would make the tool far more actionable for facility-level resource allocation and community health worker deployment planning.

A Note on What This Tool Can and Cannot Claim

The MHVI is a risk stratification index, not a mortality prediction model. It identifies which regions carry the highest concentration of known maternal health risk factors — it does not directly predict how many women will die in any given region. Policy decisions should use it alongside other evidence sources, including facility-level data, community health worker reports, and maternal death audit findings.

The tool and all underlying data and code are open-source and freely available at github.com/emmanuella-danso. Any regional health directorate, NGO, or researcher is welcome to use, adapt, or extend this work at no cost.

The data to understand maternal vulnerability in Ghana already existed. This tool makes it usable.

— End of Policy Brief —

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